

PS 657 - Cybersecurity Management & Governance - Policy Aspects
End Semester Exam – Part 1
29-04-2026

Time 60 mins

Max Marks - 25

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There are 50 multiple-choice questions. Each has only one answer. Choose the correct answer by placing a tick mark at the end of the option. No cancellation is permitted. Each question carries 0.5 marks.

1. Which of the following is an example of two-factor authentication?
 - a) Password only
 - ☒ b) Password + OTP
 - c) Username + password
 - d) PIN only
2. Biometric authentication relies on:
 - a) Something you know
 - b) Something you have
 - ☒ c) Something you are
 - d) Something you do
3. Which protocol is commonly used for Single Sign-On (SSO)?
 - a) SMTP
 - ☒ b) SAML
 - c) FTP
 - d) SNMP
4. Multi-factor authentication improves security by:
 - a) Using longer passwords
 - ☒ b) Combining different types of factors
 - c) Encrypting all data
 - d) Reducing login attempts
5. Which of the following is NOT a common authentication factor?
 - a) Knowledge (passwords)

- b) Possession (tokens)
 - c) Inheritance (biometrics)
 - ✓d) Compression (data size)
6. Which access control model is based on roles assigned to users?
- a) DAC
 - ✓b) RBAC
 - c) MAC
 - d) ABAC
7. In discretionary access control (DAC), access decisions are made by:
- a) System administrator
 - ✓b) Resource owner
 - c) Security policy
 - d) Role assignment
8. Mandatory Access Control (MAC) is typically used in:
- ✓a) Military and government systems
 - b) E-commerce websites
 - c) Social media platforms
 - d) Personal devices
9. Attribute-Based Access Control (ABAC) makes decisions based on:
- a) User roles only
 - ✓b) Attributes of users, resources, and environment
 - c) Password strength
 - d) Encryption algorithms
10. The principle of least privilege means:
- a) Users should have maximum access rights
 - ✓b) Users should only have access necessary for their tasks
 - c) Access is granted based on seniority
 - d) All users share the same access level
11. Identity federation allows:
- a) Multiple users to share one account
 - ✓b) Users to access resources across different domains with one identity
 - c) Encryption of user credentials
 - d) Automatic password resets

12. Which of the following is a risk in IAM if not properly managed?

- a) Strong authentication ✗
- b) Privilege escalation
- ✓ c) Role-based access control
- d) Multi-factor authentication ✗

13. De-provisioning in IAM ensures:

- a) Users gain more privileges
- ✓ b) Access rights are revoked when no longer needed
- c) Passwords are automatically reset
- d) Encryption keys are distributed

14. What is the main disadvantage of symmetric key cryptography?

- a) Slow encryption speed
- ✓ b) Key distribution problem
- c) Requires large key sizes
- d) Cannot encrypt large files

15. In symmetric encryption, both sender and receiver must:

- a) Use different keys
- ✓ b) Share the same key
- c) Use public keys only
- d) Exchange certificates

16. Which of the following is an asymmetric key algorithm?

- a) AES
- ✓ b) RSA
- c) DES
- d) Blowfish

17. Asymmetric cryptography is primarily used for:

- a) Fast bulk data encryption
- ✓ b) Key exchange and digital signatures
- c) Compressing data
- d) Generating random numbers

18. In RSA, the public key is used for:

- a) Decryption only
- ✓ b) Encryption and signature verification

- c) Key generation
 - d) Hashing
19. The main advantage of asymmetric cryptography over symmetric cryptography is:
- a) Faster performance
 - ☒ b) No need to share secret keys
 - c) Smaller key sizes
 - d) Easier implementation
20. Which of the following is a property of a secure hash function?
- a) Reversibility
 - b) Collision resistance
 - c) Symmetry
 - ☒ d) Key dependency
21. SHA-256 produces an output of:
- a) 128 bits
 - ☒ b) 256 bits
 - c) 512 bits
 - d) Variable length
22. Hash functions are commonly used in:
- a) Key exchange
 - ☒ b) Digital signatures
 - c) Random number generation
 - d) Compression
23. Public Key Infrastructure (PKI) is mainly used for:
- a) Generating random numbers
 - ☒ b) Managing digital certificates and keys
 - c) Encrypting bulk data
 - d) Compressing files
24. In key management, a session key is:
- a) A permanent key used for all communications
 - ☒ b) A temporary key used for one communication session
 - c) A public key stored in PKI
 - ? → d) A master key used for generating other keys

25. Which of the following ensures secure storage of cryptographic keys?

- a) Hash functions
- ☒ b) Hardware Security Modules (HSMs)
- c) Digital signatures
- d) Compression algorithms

26. Which device is primarily used to connect different networks together?

A. Switch ☒

☒ B. Router

C. Hub

D. Repeater ☒

27. What does DNS stand for?

- ☒ A. Domain Name System
- B. Data Network Service
- C. Digital Node Server
- D. Dynamic Network Setup

28. Which layer of the OSI model is responsible for routing?

- A. Data Link Layer
- ☒ B. Network Layer
- C. Transport Layer
- D. Application Layer

29. Which of the following is a private IP address?

- ☒ A. 192.168.1.1
- B. 8.8.8.8
- C. 172.217.0.46
- D. 54.23.12.5

30. What type of attack involves overwhelming a system with traffic?

- A. Phishing
- B. SQL Injection
- ☒ C. DoS/DDoS
- D. Man-in-the-Middle

31. What does a firewall primarily do?

- A. Encrypts data
- ☒ B. Blocks unauthorised access

- C. Stores passwords
 - D. Manages bandwidth
32. Which protocol is used for secure web browsing?
- A. HTTP
 - ☒ B. HTTPS
 - C. FTP
 - D. Telnet
33. Which of the following is NOT a wireless security protocol?
- A. WPA2
 - B. WEP
 - ☒ C. SSL
 - D. WPA3
34. What is the main purpose of a VPN?
- A. Increase bandwidth ☒
 - ☒ B. Provide secure remote access
 - ☒ C. Block malware
 - D. Speed up browsing ☒
35. Which tool is commonly used for network packet analysis?
- ☒ A. Wireshark
 - B. Nmap
 - C. Metasploit
 - D. Nessus
36. Which of the following is a strong password policy?
- A. Using only lowercase letters
 - B. Short passwords for convenience
 - ☒ C. Combination of letters, numbers, and symbols
 - D. Reusing old passwords
37. Which attack exploits vulnerabilities in SQL databases?
- A. Phishing
 - ☒ B. SQL Injection
 - C. Cross-Site Scripting
 - D. Buffer Overflow

38. Which of the following is considered a social engineering attack?

- ☒ A. Phishing
- B. DDoS
- C. Malware injection
- D. Port scanning

39. Which of the following is NOT a type of malware?

- A. Worm
- B. Trojan horse
- ☒ C. Firewall
- D. Ransomware

40. What does the term "zero-day vulnerability" mean?

- A. A bug discovered after patching
- ☒ B. A vulnerability exploited before a patch is available
- C. A virus that lasts one day
- D. A system crash caused by malware

41. Which malware disguises itself as legitimate software?

- ☒ A. Worm
- ☒ B. Trojan horse
- C. Spyware X
- D. Rootkit X

42. What is the primary purpose of ransom ware?

- A. Steal passwords
- ☒ B. Encrypt files and demand payment
- C. Monitor network traffic
- D. Delete system logs

43. Which of the following is a common sign of spyware infection?

- A. Faster system performance X
- ☒ B. Unauthorized pop-ups and data leaks
- C. Improved network speed X
- D. Automatic system updates

44. What is the main goal of digital forensics?

- A. Prevent cyber-attacks
- ☒ B. Investigate and preserve digital evidence

- C. Encrypt sensitive data
- D. Speed up system performance

45. Which malware replicates itself without user intervention?

- ☒ A. Trojan
- ☒ B. Worm
- C. Spyware X
- D. Adware X

46. Which of the following is an example of social engineering?

- A. SQL Injection
- ☒ B. Phishing email
- C. Buffer overflow
- D. Malware infection

47. What does "chain of custody" refer to in digital forensics?

- A. Encryption of evidence
- ☒ B. Documentation of evidence handling
- C. Backup of forensic data
- D. Secure deletion of files

48. Which malware records keystrokes to steal information?

- A. Worm X
- ☒ B. Keylogger
- C. Trojan X
- D. Spyware

49. Which of the following is a common forensic technique for analysing deleted files?

- ☒ A. File carving
- B. Packet sniffing
- C. Port scanning
- D. Encryption

50. Which forensic principle ensures evidence remains unchanged during investigation?

- A. Non-repudiation
- ☒ B. Integrity
- C. Confidentiality
- D. Availability

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== Short Answer Questions (3 marks each) ==

✓ **Q1.** Describe how risk can be quantified.

- 3 marks

✓ **Q2.** Describe the concept of a control as used in management and cybersecurity frameworks

- 3 marks

✓ **Q3.** What is a consent manager as described in the Indian Personal Data Protection Act 2023? – 3 marks

✓ **Q4.** Describe the need for encryption in digital transactions and the tension with law enforcement

- 3 marks

✓ **Q5.** What are the ways in which an organisation can arrive at an appropriate investment in cybersecurity?

- 3 marks

===== Essay type question =====

Q6. How would you assess the readiness of an organization like IIT Bombay to recover from a ransomware attack? If such an attack actually did take place describe in details the steps that are needed to deal with it and restore normal operation. - 10 marks